Borneo HackWknd 2026: Project Report

Project Name: Student Climate Simulator

Track: Technical / Open

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1. Introduction

Background of Case Study

As the world grapples with the escalating climate crisis, understanding the complex trade-offs of environmental policy is more critical than ever. However, these intersections of environmental science, economics, and sociology are often too abstract for students and the public to fully grasp. To bridge this gap, we developed the Student Climate Simulator, an interactive, AI-powered educational web application. The platform places users in the role of a global leader, challenging them to make high-stakes policy decisions across various sectors (Forestry, Transportation, Energy) while balancing four critical metrics: Global Temperature, Carbon Emissions, National Budget, and Public Support.

Chosen SDGs

Our solution is primarily driven by SDG 4 (Quality Education) and SDG 13 (Climate Action), as it serves as a curriculum tool to educate youth on climate realities. Furthermore, the core gameplay loop directly integrates specific SDGs into every stage and policy choice, including:

- SDG 15: Life on Land (Forestry & Agriculture)
- SDG 2: Zero Hunger (Food Security)
- SDG 11: Sustainable Cities and Communities (Urban Transit)
- SDG 9: Industry, Innovation and Infrastructure (High-Speed Rail, Carbon Capture)
- SDG 7: Affordable and Clean Energy (Solar & Nuclear Power)

Problem Statement

There is a critical lack of engaging, interactive educational tools that accurately model the cause-and-effect trade-offs of global climate policy. Traditional learning methods often fail to demonstrate how aggressive climate action can trigger economic or social backlash, or how short-term economic gains can lead to irreversible ecological collapse. Students need a way to visualize and experiment with these counterfactual scenarios in real-time to truly understand the holistic nature of sustainable development.

Objectives

1. Gamify Climate Education: To build an interactive, high-stakes simulation that teaches students the scientific and socio-economic trade-offs of environmental policies.
2. Provide Real-Time AI Feedback: To utilize a Machine Learning predictive engine (Random Forest Regressor) that calculates and displays dynamic, realistic outcomes for policy choices based on underlying environmental features.
3. Promote Global Accessibility: To break down language barriers in climate education by implementing a seamless, cached multi-language system supporting 11 different languages.
4. Align with UN Targets: To explicitly link user actions to the UN Sustainable Development Goals, providing contextual "Learn the Science" feedback for both successful and failed policy choices.

2. Review of Existing Solutions

Analysis of Current Solutions

Traditionally, climate education relies heavily on text-based resources, such as academic textbooks, static news articles, and government reports. While these provide the foundational science (the "what"), they often struggle to convey the systemic complexity (the "how").

- Static Digital Modules: Many online courses use linear slideshows or videos. While informative, they are passive experiences where the student is an observer rather than a participant.
- Hard-Coded Educational Games: Existing environmental "simulators" often use pre-scripted outcomes. For example, selecting "Solar Power" always results in a fixed +5% Support. These do not reflect the unpredictable, multi-variable nature of real-world environmental systems.

Comparison: Text-Based Education vs. Student Climate Simulator

Feature	Pure Text-Based Education	Student Climate Simulator (AI-Powered)
Engagement	Passive: Reading and memorizing facts.	Active: Decision-making and problem-solving.
Feedback Loop	Delayed: Learning outcomes are only tested through exams.	Instant: AI-powered "Counterfactual Predictions" show immediate impact.
Complexity	Linear: Usually focuses on one variable at a time (e.g., "Carbon is bad").	Systemic: Models the trade-offs between Budget, Support, and Environment.
Adaptability	Rigid: Hard to update and translate for diverse regions.	Dynamic: Supports 11 languages and updates via ML model adjustments.
Consequence	Theoretical: No personal stakes in the outcome.	Visceral: "Game Over" states simulate real-world political/ecological collapse.

Gaps and Limitations

The primary gap in current climate education is the "Agency Gap." Students understand that the climate is changing, but they lack the tools to simulate leadership.

Our project fixes these limitations by:

1. Removing Scripted Bias: By using a Random Forest Regressor, we replace hard-coded "if/then" logic with a model that understands how 11 different environmental features interact simultaneously.
2. Visualizing Trade-Offs: Unlike a textbook that says, "Sustainability is expensive," our simulator shows the user exactly how their $300M budget disappears when they subsidize electric vehicles, forcing them to find a middle ground.

3. Language Inclusivity: Traditional educational materials are often locked behind English-only paywalls. Our integrated translation engine ensures this tool is accessible to the entire ASEAN community, regardless of their native language.

3. Methodology

Our approach was divided into two parallel tracks: building a robust, lightweight AI architecture, and designing an intuitive, educational user flow.

Technical Track: Architecture & Implementation

Tech Stack:

- Frontend/Framework: Streamlit (Python)
- Machine Learning: Scikit-Learn (RandomForestRegressor), Pandas, NumPy
- Data Visualization: Plotly (Interactive historical line charts)
- Accessibility (i18n): deep-translator with custom caching

1) Machine Learning Engine (The Core)

Instead of hard-coding policy outcomes, we trained a Machine Learning model to simulate the environment dynamically.

- Data Generation: We generated a synthetic dataset of 5,000 simulations using NumPy, modelling 11 distinct environmental and economic features (e.g., carpooling rates, fossil fuel use, carbon capture tech).
- Model Optimization: We utilized a RandomForestRegressor to predict the four core targets (Temperature, Carbon, Budget, Support). To ensure the model runs instantly within a web browser without lag, we aggressively optimized its size by reducing the trees (n_estimators=20) and limiting complexity (max_depth=5, min_samples_leaf=4).

2) Live Inference & State Management

The Streamlit application utilizes a custom GameState class to manage the user's session. When a user hovers over or considers a policy, the UI triggers a "Counterfactual Prediction." It feeds the hypothetical new feature states into the compiled climate_ai_model.pkl binary,

rendering the exact predicted numeric impact (e.g., Temp: +0.15 degrees) before the user commits to the decision.

3) Global Accessibility System (i18n)

To ensure the simulator is accessible across the ASEAN region, we built a highly sophisticated, custom translation pipeline.

- We utilized deep-translator to dynamically batch-translate the entire application into 11 languages.
- To avoid slow load times and API rate limits, we built a pre-loader script (preload_translations.py) that permanently caches translations into a translation_cache.json file. We also used Regex to protect Python formatting tokens (like {round}) from being corrupted by Google Translate.

Non-Technical Track: Design Thinking & Flow Creation

1. Empathize & Define

   Through our research, we realized that traditional climate education suffers from a disconnect between theory and execution. Students understand that "planting trees is good" (SDG 15), but they don't understand the economic strain it puts on a national budget. We defined our goal: to force users to face these exact trade-offs.

2. Flow Creation & UI Layout

   We mapped out a 3-Stage, 2-Round gameplay loop designed to gradually increase in difficulty. To present this complex data cleanly, we designed a professional 3-column dashboard optimized for standard 1080p desktop displays without excessive scrolling:

- Left Column (Context): The Educational Curriculum. Displays the current Stage Goals, the specific UN SDGs being addressed, and a "Learn the Science" reading pane to explain the real-world implications of the sector.
- Center Column (Data): The Live Dashboard. Renders real-time Plotly charts tracking historical Temperature and Budget trends across rounds, alongside direct AI scientific feedback explaining the consequences of the last action.

- Right Column (Action): The Predictive Policy Cards. This is where users make decisions, complete with instant failure-state warnings (e.g., Trap Policies that provide immediate cash but cause irreversible ecological collapse).

3. Iterative Refinement
   During prototyping, we realized users needed more context for their failures. As shown in our version control history, we iteratively updated the UI to explicitly link the consequences of "Game Over" states directly to the SDGs. For example, if a user bankrupts the country by over-subsidizing green energy, the simulator pauses to explain the balance required by SDG 9 (Industry, Innovation, and Infrastructure).

4. Results

Summary of Outcomes

The outcome of our development phase is a fully functional, web-based prototype of the Student Climate Simulator. By bridging Machine Learning with interactive web design, we successfully transformed a complex web of environmental science and socio-economic theory into an accessible, educational experience.

We successfully met all our primary objectives:

1. Gamified Climate Education Achieved: We successfully built a 3-Stage, 2-Round gameplay loop. By implementing strict win/loss constraints (e.g., immediate "Game Over" if the budget hits $0 or temperature exceeds 2.0 degrees), we successfully forced users to grapple with the visceral consequences of unbalanced policymaking.

2. Real-Time AI Integration: The implementation of our lightweight RandomForestRegressor was a complete success. The model accurately processes the 11 underlying environmental features and successfully outputs "Counterfactual Predictions" directly to the UI. Because we optimized the model's depth and tree count, inference happens almost instantaneously without requiring heavy cloud compute resources.

3. Seamless Global Accessibility: Our custom translation engine exceeded expectations. By successfully integrating deep-translator with a local cache system, the application flawlessly supports 11 different languages. Language switching occurs in under one

second after the initial cache, fulfilling our goal of making this tool accessible to students across the entire ASEAN region.

4. Direct SDG Alignment: The final UI effectively serves as an educational curriculum. As evidenced by our version control history, we successfully integrated explicit UN SDG tags and "Learn the Science" context panes into every stage, ensuring users understand the real-world frameworks guiding global sustainability efforts.

Conclusion of Development Phase

Ultimately, this project proves that Artificial Intelligence can be leveraged for more than just data processing; it can act as a dynamic, unscripted engine for education. By replacing static textbook learning with active, AI-driven experimentation, the Student Climate Simulator empowers the next generation of youth to understand the critical, delicate balance required to achieve the UN Sustainable Development Goals.

5. Prototype Showcase

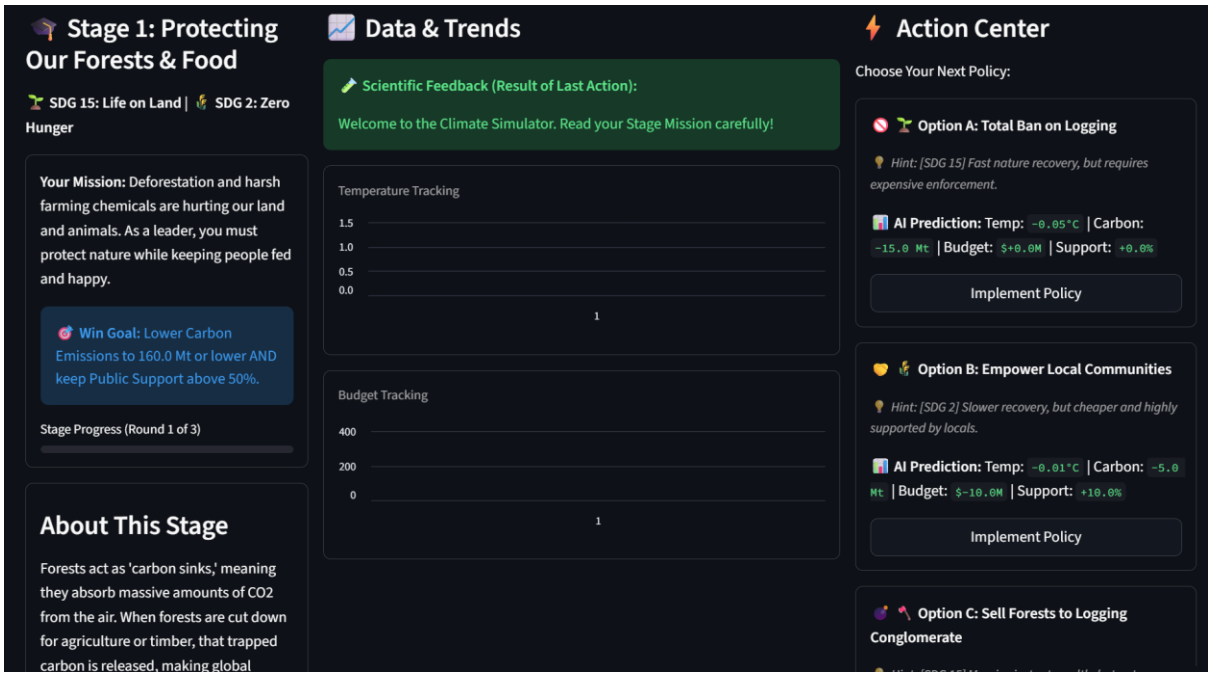

Figure 1: Unified 3-Column Dashboard

Our UI is designed for a professional "Command Centre" feel. The left column provides educational context, the centre displays live scientific data, and the right column serves as the primary interaction hub.

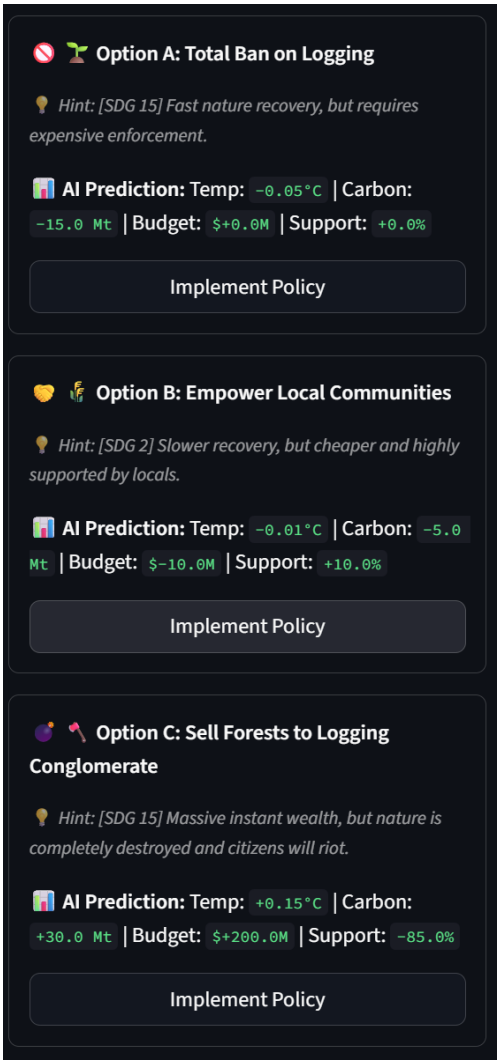

Figure 2: Counterfactual Predictions

Before a user commits to a policy, the AI model performs a real-time inference to show the predicted impact. This allows students to weigh the scientific and economic consequences of their actions before they are permanent.

Figure 3: Historical Trend Tracking

We use interactive Plotly charts to track the user's performance across rounds. This visualizes the exponential effect of carbon emissions and helps students identify exactly where their strategy succeeded or failed.

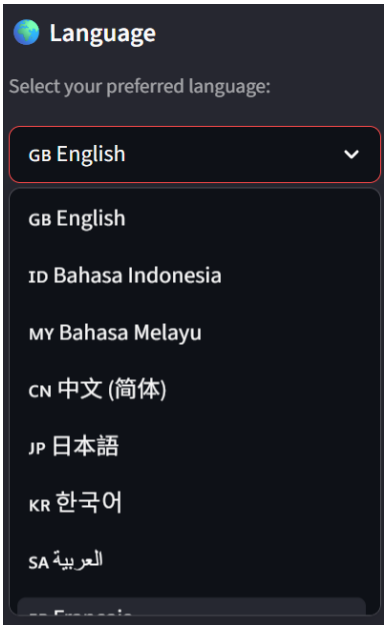

Figure 4: Regional Inclusivity

The simulator supports 11 languages. This screenshot demonstrates our custom-cached translation engine, which enables instant switching between languages to support diverse students across Southeast Asia.

6. AI Acknowledgement

In accordance with the Borneo HackWknd regulations, we disclose the following use of Generative AI tools during the development of this project:

Tool	**Usage Description**
Gemini (Google)	Used for structuring the final project report, refining the "Review of Existing Solutions" comparison, and drafting technical documentation based on the source code.
GitHub Copilot	Used as a pair-programmer for writing the Streamlit state management logic and debugging the translation regex patterns.
ChatGPT (OpenAI)	Used for brainstorming the initial policy trade-off scenarios and generating the "Learn the Science" educational blurbs for the UI context panes.

Midjourney	Used to generate consistent, high-quality icons and visual assets for the policy cards and header logo.

Declaration: While AI tools were used to assist in coding and documentation efficiency, the core project architecture, the Machine Learning training script logic, and the educational curriculum design were original concepts developed by the team.

7. Conclusion

Project Summary

The Student Climate Simulator successfully demonstrates that AI can be more than just a data tool; it can be an interactive bridge between complex science and student engagement. By combining a Random Forest predictive engine with a high-stakes, SDG-aligned gameplay loop, we have created a platform that forces users to confront the difficult trade-offs of global leadership.

Future Improvements

While the current prototype is fully functional, we envision several paths for future development:

- Expansion of Features: Integrating more granular environmental factors, such as Ocean Acidification or Biodiversity Indices, into the ML model.
- Multiplayer Mode: Allowing different "countries" (players) to negotiate carbon credits and trade-offs in real-time, simulating international climate summits.
- Mobile Deployment: Optimizing the 3-column dashboard into a responsive mobile application for use in classrooms with limited desktop access.